11.16.3.21.1

EE24BTECH11013 - MANIKANTA D

Question:

In a class of 60 students, 30 opted for NCC, 32 opted for NSS, and 24 opted for both NCC and NSS. If one of these students is selected at random, find the probability that:

1) The student opted for NCC or NSS.

1 Theoretical Solution

The total number of students in the class is:

$$|S| = 60. \tag{1.1}$$

The number of students who opted for NCC or NSS can be calculated using the formula for the union of two sets:

$$|A \cup B| = |A| + |B| - |A \cap B|. \tag{1.2}$$

where:

$$|A| = 30, \quad \text{(students who opted for NCC)} \tag{1.3}$$
$$|B| = 32, \quad \text{(students who opted for NSS)} \tag{1.4}$$
$$|A \cap B| = 24, \quad \text{(students who opted for both NCC and NSS).} \tag{1.5}$$

Substituting these values, we get:

$$|A \cup B| = 30 + 32 - 24 = 38. \tag{1.6}$$

The probability of a student opting for NCC or NSS is:

$$P(A \cup B) = \frac{|A \cup B|}{|S|} = \frac{38}{60} = \frac{19}{30} \approx 0.6333. \tag{1.7}$$

2 Event Definition

Define events as shown in Table **??**:

Event	Denotation
A'	Student does not opt for NCC
A	Student opts for NCC
B'	Student does not opt for NSS
B	Student opts for NSS

TABLE 1: Defining Events

Boolean Expression	Interpretation
$A + A' = 1$	Complement Law
$AB + A'B = B$	Absorption Law
$AB + AB' = A$	Distributive Law
$A + B = AB + AB' + A'B$	Union of Sets
$P(A + B) = P(A) + P(B) - P(AB)$	Inclusion-Exclusion Principle
$P(A'B') = 1 - P(A + B)$	Complement Rule

TABLE 1: Boolean Algebra Rules

3 Boolean Algebra Principles

Below are some axioms and theorems from Boolean algebra:
For any two event A and B,

$$\because A + A' = 1 \tag{1.8}$$

$$AB + A'B = B \tag{1.9}$$

$$\implies \Pr(AB) + \Pr(A'B) = \Pr(B) \tag{1.10}$$

$$\because B + B' = 1 \tag{1.11}$$

$$AB + AB' = A \tag{1.12}$$

$$\implies \Pr(AB) + \Pr(AB') = \Pr(A) \tag{1.13}$$

$$A + B = AB + AB + AB' + A'B \tag{1.14}$$

$$A + B = AB + AB' + A'B \tag{1.15}$$

$$\Pr(A + B) = \Pr(AB) + \Pr(AB') + \Pr(A'B) \tag{1.16}$$

Adding (1.10),(1.13) and (1.17) and cancelling same terms

$$\Pr(AB) = \Pr(A) + \Pr(B) - \Pr(A + B) \tag{1.17}$$

$$\because \Pr(A'B') = \Pr((A + B)') \tag{1.18}$$

$$\Pr(A'B') = 1 - \Pr(A + B) \tag{1.19}$$

$$P(A) = \frac{30}{60}, \quad P(B) = \frac{32}{60}, \quad P(AB) = \frac{24}{60}. \tag{1.20}$$

The fundamental axioms of probability are as follows:

Non-Negativity Axiom:

$$P(A) \geq 0$$

Normalization Axiom:

$$P(S) = 1$$

Additivity Axiom:

If $A_1, A_2, A_3, \ldots$ are mutually exclusive (disjoint) events, then:

$$P(A_1 \cup A_2 \cup A_3 \cup \ldots) = P(A_1) + P(A_2) + P(A_3) + \ldots$$

From the given data:

$$P(A) = \frac{30}{60}, \quad P(B) = \frac{32}{60}, \quad P(AB) = \frac{24}{60} \tag{1.21}$$

Now applying probability axioms:

$$P(A + B) = P(A) + P(B) - P(AB) \tag{1.22}$$

$$= \frac{30}{60} + \frac{32}{60} - \frac{24}{60} \tag{1.23}$$

$$= \frac{38}{60} = \frac{19}{30} = 0.6333 \tag{1.24}$$

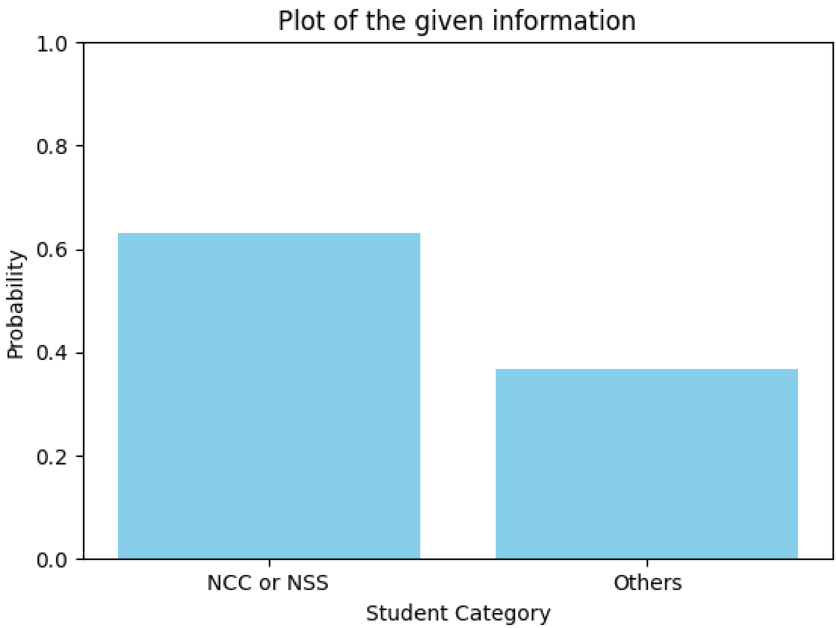

Fig. 1.1: Probability of opting for NCC or NSS using Boolean logic